A
Practical Activity Report
Submitted for

DATA ENGINEERING-(UCS677)

END-Semester Lab Evaluation

Social Media Website

Submitted to-

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BE Third Year

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INTRODUCTION

Social media has transformed the way individuals connect, communicate, and collaborate, especially within academic communities. With the rapid advancement of technology, there is a growing need for user-friendly, efficient, and scalable platforms tailored to students' needs. This project, titled **"TIET Social – A College Social Media Website Using React.js, MongoDB, Express, and Node.js,"** aims to bridge the gap between student interaction and digital connectivity by leveraging modern web technologies.

Built using the MERN stack, the project delivers a feature-rich platform that simplifies and enhances online social engagement. The front end, powered by React.js, ensures a dynamic and responsive user interface, providing an intuitive and interactive experience. The back end, developed with Node.js and Express, efficiently manages server-side operations and seamless communication between client and server. MongoDB serves as the database, offering a scalable and flexible solution for storing and managing user profiles, posts, messages, and notifications.

The platform incorporates essential social media functionalities such as user authentication, profile creation, posting updates, liking and commenting on posts, and real-time messaging. Additionally, it includes administrative capabilities for monitoring platform activities, ensuring a safe and engaging environment for all users. Designed with scalability and performance in mind, the application ensures compatibility across devices while maintaining a secure environment for handling user data.

By addressing the challenges faced by students in building strong digital connections within a college ecosystem, this project serves as a model for developing modern, efficient, and engaging social media platforms. It highlights the integration of contemporary technologies to create a vibrant, robust, and student-focused online community.

PROBLEM STATEMENT

The increasing digitization of communication has made social media an essential part of student life, offering opportunities to connect, collaborate, and share experiences beyond physical boundaries. However, many existing social platforms are either too broad, lacking the specificity needed for college communities, or overly complex, making them less appealing for focused student interaction. Students at TIET currently lack a dedicated, campus-specific digital space where they can easily engage with peers, share updates, organize events, and build stronger community ties.

Moreover, students today expect seamless, responsive, and intuitive digital experiences. A poorly designed platform can result in low engagement, communication gaps, and reduced participation in campus life. Key features such as real-time updates, event announcements, direct messaging, and user privacy controls are often missing or underdeveloped in generic platforms, creating a gap between what students expect and what is available.

Another critical challenge lies in platform management and scalability. As student participation grows, administrators need efficient tools to manage users, monitor content, and ensure a safe and vibrant online environment. Without a robust and flexible system, maintaining platform security, user data privacy, and content moderation becomes increasingly difficult.

This project aims to address these challenges by developing a scalable, user-friendly, and student-centric social media platform specifically for TIET. Built using the MERN stack (MongoDB, Express, React.js, and Node.js), the solution integrates modern technologies to offer a seamless and engaging experience for students while simplifying administrative oversight. By focusing on relevance, ease of use, scalability, and community-building, **TIET Social** bridges the gap between traditional college interactions and the demands of the modern digital student community.

SPECIFIC REQUIREMENTS

a) Functional Requirements

i) **User Authentication & Authorization:**
Users must be able to register, log in, and log out securely. This includes features such as password encryption, account verification, and session management.

ii) **Profile Management:**
Users should have the ability to create and edit their profiles, including adding profile pictures and updating personal information.

iii) **Posting Functionality:**
Users should be able to create, edit, and delete posts. Posts may include text, images, or other multimedia content. Additionally, users should have the option to like, comment on, and share posts.

iv) **Friend Management:**
Users should be able to search for other users, send friend requests, accept or reject friend requests, and manage their list of friends.

v) **Photo Upload:**
Users should have the ability to upload photos to their profiles or as part of their posts. The application should support image storage, resizing, and display.

vi) **Scrolling:**
Smooth scrolling functionality is implemented throughout the application to enhance user experience, particularly when navigating through long lists of content.

b) Non-Functional Requirements

i) Performance:
The application should remain responsive and performant, even under heavy user activity, such as during peak posting and interaction times. Page load times and server response times must be minimized to ensure a smooth and engaging user experience.

ii)Scalability:
The architecture of TIET Social is designed to scale horizontally to accommodate an increasing number of users, posts, comments, and media content as the college community grows.

iii)Security:
Strong security measures will be implemented to protect user data and privacy. This includes secure storage of passwords (using encryption), encryption of sensitive user information, and protection against common security threats such as Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and SQL Injection.

iv)Usability:
The user interface will be intuitive and easy to navigate, with clearly labeled elements and consistent design patterns. Regular user feedback will be incorporated to continually enhance usability and user satisfaction.

v)Accessibility:
TIET Social will be designed to be accessible to users with disabilities, adhering to Web Content Accessibility Guidelines (WCAG), ensuring an inclusive platform for all students.

vi)Compatibility:
The application must be compatible across a wide range of devices (desktops, tablets, smartphones) and major web browsers such as Chrome, Firefox, Safari, and Edge to ensure accessibility for all users.

vii)Maintainability:
The codebase will be clean, modular, and well-documented, making it easy to maintain, debug, and extend with future updates or feature enhancements.

viii)Reliability:
TIET Social should be highly reliable and available at all times, with minimal downtime for maintenance or upgrades. The system should handle errors gracefully and provide clear, user-friendly error messages when necessary.

CONTEXT LEVEL AND DATA FLOW DIAGRAM

Context Level Diagram -(Level 0)

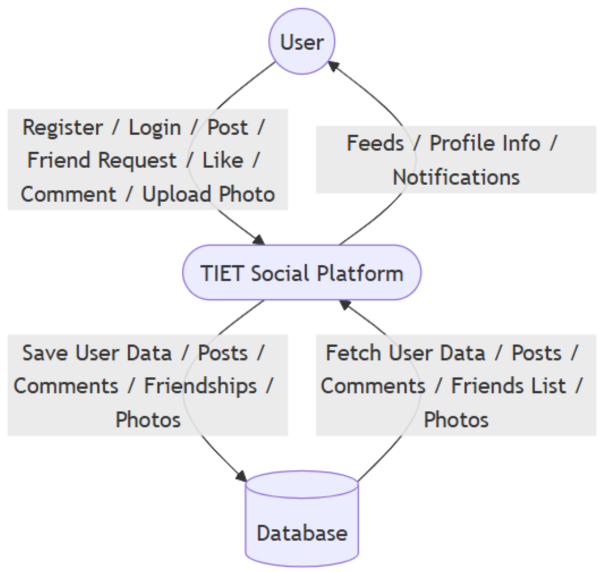


Fig 1.

Data Flow Diagram -(Level 1)

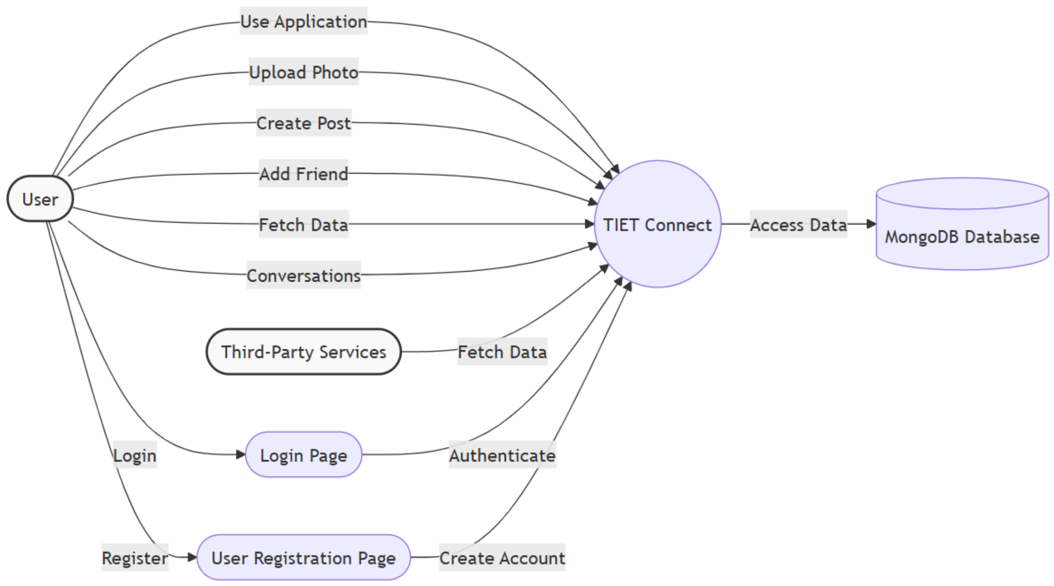


Fig 2.

SYSTEM SPECIFICATION

1. Hardware Specifications

Development Environment:

- **Processor**: Intel Core i5/i7 or AMD Ryzen 5/7 (or equivalent).
- **RAM:** 8 GB (minimum); 16 GB (recommended for smoother development).
- **Storage:** 256 GB SSD (minimum); 512 GB SSD or higher (recommended).
- **Graphics:** Integrated GPU (sufficient for web development).
- **Display**: Full HD (1920x1080) resolution monitor.

2. Software Specifications:

Development Tools:

- **Operating System**: Windows 11.
- **IDE/Editor**: Visual Studio Code.
- **Version Control**: Git with GitHub for code management.
- **Package Manager**: npm (Node Package Manager).

Frameworks and Libraries:

- **Frontend**: React.js with Bootstrap for styling.
- **Backend**: Node.js with Express.js for API development.
- **Database**: MongoDB(NoSQL database for efficient data handling).

Browsers:

- Google Chrome.

Additional Tools:

- **MongoDB Atlas**: GUI for managing MongoDB.

TOOLS USED

The successful implementation of the Full Stack Social Media Website was made possible by leveraging a variety of tools for development, testing, and deployment. These tools were chosen to ensure efficiency, scalability, and maintainability. Below is a detailed explanation of the tools used and their roles in the project:

1. MERN Stack

- **MongoDB**: A NoSQL database used to store and retrieve product details, user information, and order data efficiently. Its schema-less nature made it adaptable to changing data structures.
- **Express.js**: A lightweight and flexible framework for building server-side applications and RESTful APIs. It helped in managing user requests and performing CRUD operations.
- **React.js**: A powerful front-end library used to create a dynamic and responsive user interface. It ensured seamless navigation and improved user experience with reusable components.
- **Node.js**: A runtime environment that allowed us to execute JavaScript on the server side. It facilitated efficient handling of back-end operations and server-side scripting.

2. Socket.IO

- Socket.IO was used to implement real-time chat functionality, enabling instant message delivery between users without page refreshes. It facilitated seamless communication by handling events like message sending and receiving, ensuring a smooth and responsive chat experience on the platform.

3. MongoDB Atlas

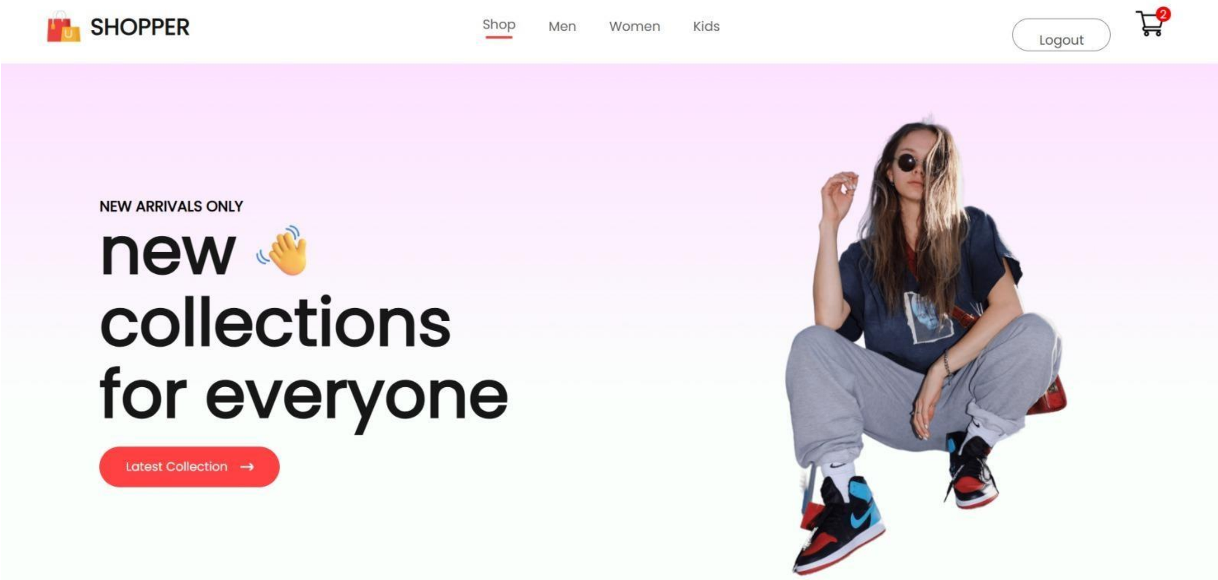
- A GUI tool for visualizing and managing the MongoDB database. It simplified querying and testing database operations during development.

4. Git and GitHub

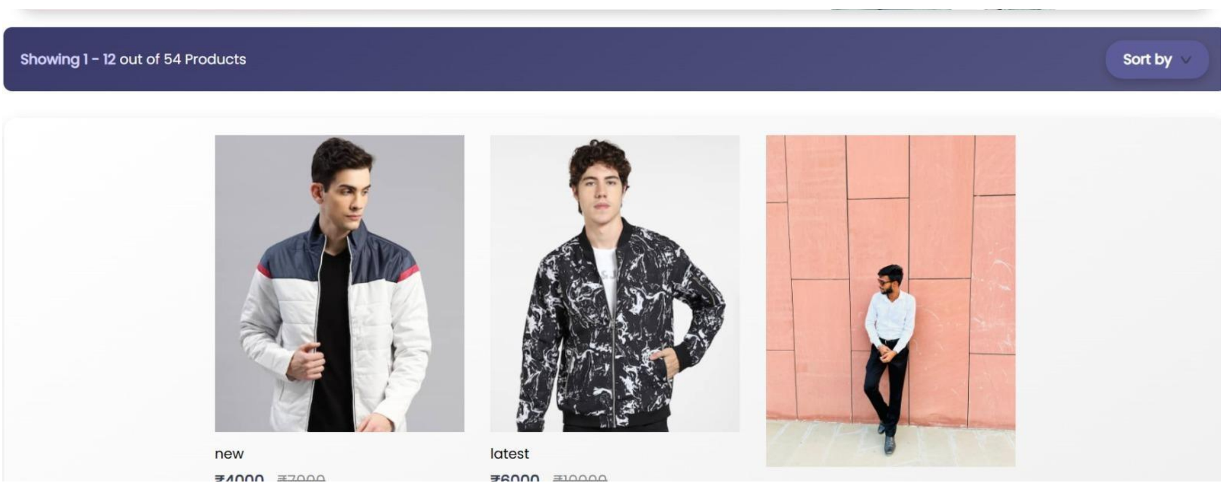
- Version control systems were employed to track changes, manage the codebase, and enable collaboration. GitHub repositories were used to store and share the project securely.

SAMPLE SCREENSHOTS

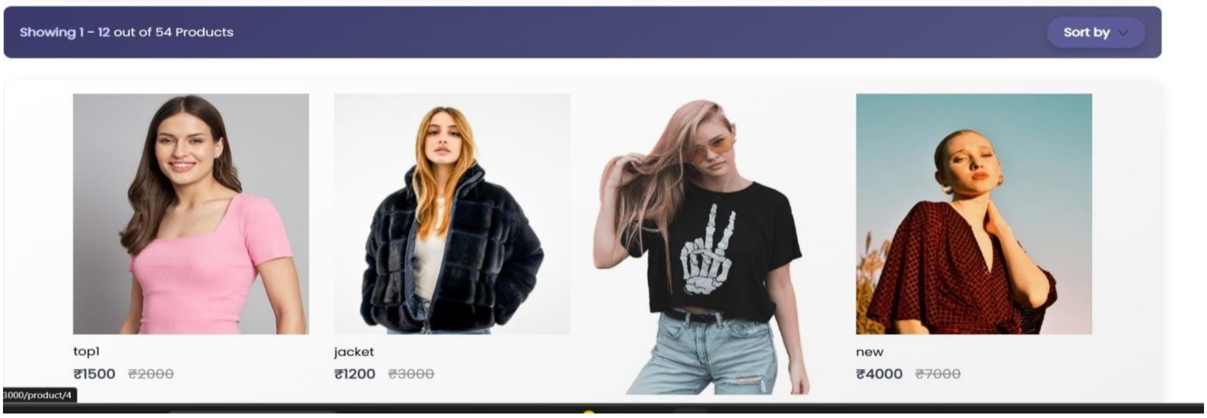
1) Main page



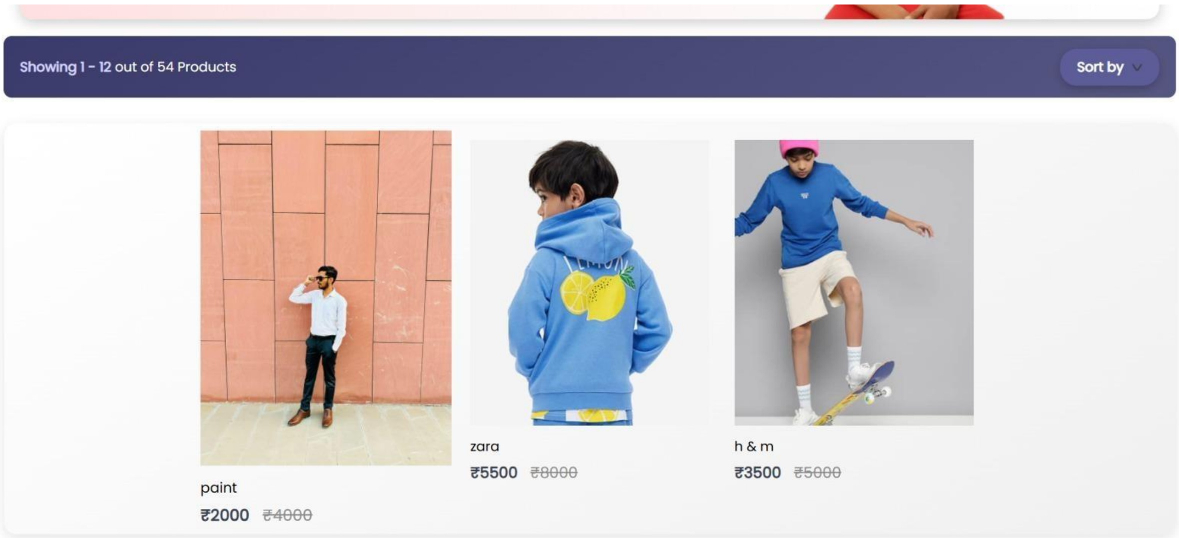
2) Men page -



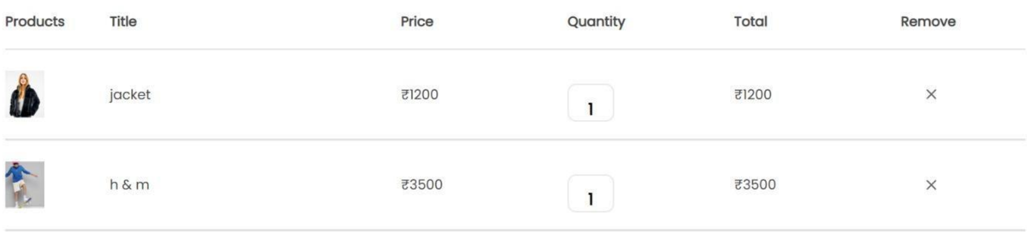
3) Women page -



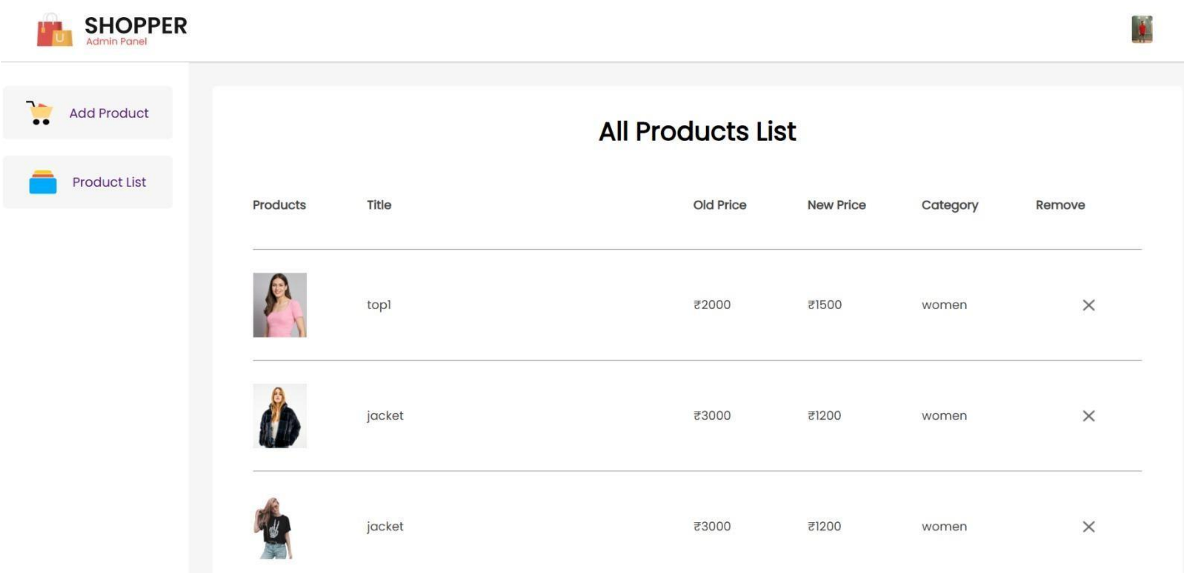
4) **Kids page -**



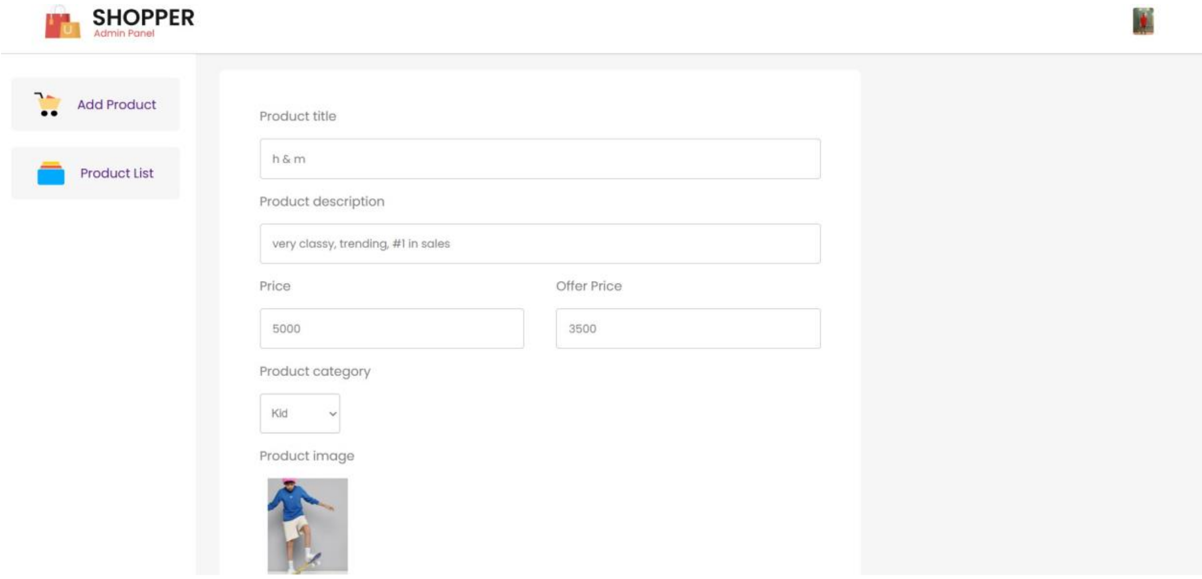
5) **Cart image -**



6) **Product List image -**



7) **Admin page image -**



OUTPUT REPORTS

This section outlines the key reports generated during the development and testing phases of the Full Stack Social Media Website. Although the actual documents are not included, the following details summarize the process and key insights.

a. API Testing Report

API testing was conducted using Postman to validate the functionality and reliability of various endpoints. The following key endpoints were tested:

- **User Login Endpoint**: Ensured successful authentication and token generation for valid credentials.
- **Product Search Endpoint**: Verified that search queries return accurate product details based on filters and categories.
- **Cart Update Endpoint**: Confirmed the addition and removal of items from the user cart, along with proper error handling for invalid requests.

b. Debugging Logs

During development, debugging was performed extensively using the following methods:

- **Console Logs**: Key application states, such as database connection status and API request payloads, were logged.
- **Error Handling Middleware**: Error messages and stack traces were captured to identify and resolve issues effectively.

CONCLUSION

The development of the Full Stack E-Commerce Website using the MERN stack has been a comprehensive journey encompassing multiple aspects of modern web application development. This project was undertaken with the aim of creating a user-friendly, efficient, and scalable platform to streamline the online shopping experience for both customers and administrators. By integrating technologies like MongoDB, Express.js, React.js, and Node.js, the system ensures seamless interactions between the front-end, back-end, and database layers.

One of the major accomplishments of this project is the dynamic and responsive interface built with React.js, enabling customers to browse and interact with the platform effortlessly. The use of MongoDB as a NoSQL database has ensured efficient storage and retrieval of data, making the system suitable for handling real-time operations like user authentication, product browsing, and order processing. Additionally, the back-end, powered by Node.js and Express.js, has been optimized for smooth API communication, ensuring that the data flow between the client and the server remains robust and secure.

The project also emphasized scalability and maintainability, making it adaptable for future enhancements such as integrating advanced features like AI-driven product recommendations or multiple payment gateways. Security has been given priority through practices such as token-based authentication (JWT), ensuring data integrity and user trust.

In conclusion, this e-commerce platform successfully demonstrates how modern web technologies can be leveraged to create a reliable, feature-rich solution. While this project has met its initial objectives, it lays a strong foundation for further development, addressing the evolving needs of both businesses and customers in the digital age. This achievement reflects not only technical proficiency but also the ability to design and implement solutions that align with real-world requirements.